* Feature Selection v/s Feature Extraction:

→ Dimensionality Reduction

1. Why Dimensionality Reduction?
   * Prevent - Curse of Dimensionality
   * Improve the performance of the model
   * Visualize the data → understand the data

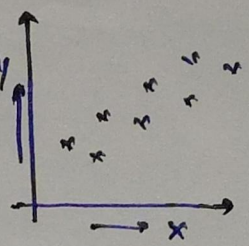
[3d] [2d]

[100d] → [3d] or [2d]

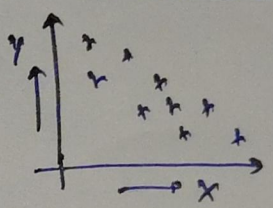
Feature Selection

I/P	O/P
x	y
–	–
–	–
–	–
–	–

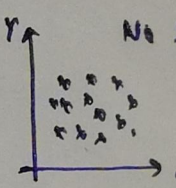
+ve ⇒ [x ↑ y ↑ ; x ↓ y ↓]



-ve ⇒ [x ↓ y ↑ ; x ↑ y ↓]



$$cov(x,y) = \sum_{i=1}^{n} \frac{(x_i - \bar{x}) * (y_i - \bar{y})}{N-1} = \begin{cases} +ve \\ -ve \\ \approx 0 \end{cases}$$



No relation b/w x & y

$$\text{Pearson correlation} = \frac{cov(x,y)}{\sigma_x * \sigma_y} = [-1 \text{ to } 1]$$

≈0 {No relation}

{ The more towards the value of +1 the more +ve correlated x & y is. } → +ve correlated

Dataset Housing

→ Independent feature

House Size	Fountain Size
–	–
–	–

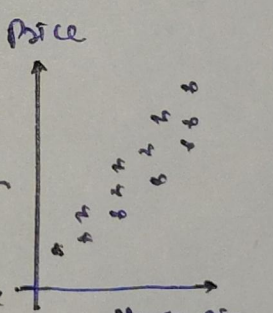
→ DROP

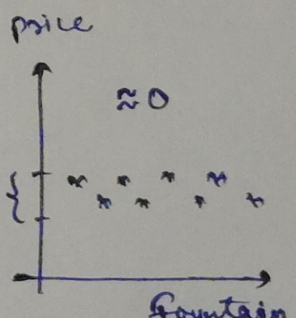
Price → o/p feature

{ Linear Relationship } or

covariance = +ve / -ve

{Feature Selection}





Feature Extraction:

Domain Expertise

Room size | No. of Rooms | [Price] [Price]

↓ ↓ ~~Transform~~

→ [House Size] | [Price]

2. Feature Extraction → 1. Feature Reduction

Transformation to extract new features.